Intelligent Learning Platform With Automated Assessment

Abstract— **The increasing dependency on digitalized academic materials and resources has made it challenging for students to keep up with the cognitive burden understanding complex materials. The proposed system introduces an AI-based study assistant designed to help relieve the cognitive overhead of extracting and learning from academic materials. This document centric system helps users from getting perplexed by varying explanations and resources. The system consists of automated summarization, semantic search, and automated assessment, all based on document context. The system uses OCR to process documents and text preprocessing to obtain clean textual data. A chat space module made using LLM, Qwen-2.5 model helps users to carry out query answering activities related to the document content. A semantic strong recommendation module using SBERT and KBERT generates keywords and runs semantic searches across the wide web and results with context relevant resource links including videos and articles via API calls. A quiz generating module built using Llama 3.3 generates quiz based strictly on the key concepts of the document content.**

Keywords— AI study assistant, LLM, Qwen-2.5, Llama 3.3, OCR, SBERT, KBERT, document processing, semantic search,

I. Introduction

In this era of the digitalised world, it seems everything around us is dependent on the internet or the online world. The education system or the management is not an exception to this. The academic institutions or the establishment have also digitalised their communication methods with the students. The students or the learners have also digitalised their learning methods, including their study materials, or referring to lectures on the internet for free. At times, one can even get their doubts clarified through the online world without anyone's help.

The recent advancements in the usage of Natural Language Processing (NLP), semantic similarity, or document understanding systems have helped in the development of intelligent learning platforms, where the systems can analyse the content or help the learners in an interactive way. There are some systems, where one can even get the contextual understanding or examples, or even develop assessments for the topic or the subject one wants to study, along with the resources provided. However, for the use of these features, students have to switch from different tabs for different tasks, i.e., they have to upload their documents and different materials again and again for different tasks. This reduces the efficiency of these AI-based learning tools for the students. We are no longer provided with a single stop platform that could allow students and learners to use all these facilities without having to switch different platforms and upload their documents again and again.

In order to overcome the limitation of the current AI-based learning tools, the proposed system provides a solution for the availability of a platform for document-centric information and contextual explanation, an interactive AI chat for asking different questions related to the content of the documents, availability of different links related to the topic of the document, of articles and videos, and a feature of assessment that strictly works on the basis of the content of the document uploaded by the students. The students will have to upload their documents only once and will be able to use all these facilities without having to switch different tabs. This platform uses Qwen 2.5 Llama 3.3, sentence transformers, and Key BERT.

II. Related works

The researchers compared and benchmarked different large language models like GPT, Qwen, and Llama to evaluate their effectiveness in generating high quality text summaries. The results showed that GPT based models generated more coherent and accurate text summaries whereas open-source models like Qwen and Llama showed competitive results with lower computational requirements. This highlighted the potential of modern LLMs for efficient text summarizations [1].

The researchers proposed a system that used prompt learning with the Qwen LLM to generate accurate for specific topic centred documents with the help of using carefully designed prompts. The results using proper topic specific prompts for guiding the Qwen models showed better semantic relevance for generating document summaries [2].

The proposed system uses phrase level text extractions to analyse article content and recommend content relevant articles based on semantic similarity. The phrase level representations boosted recommendation accuracy and relevance when compared the traditional word level-based recommendation system [3].

The researchers proposed a system that combines Sentence-BERT for semantic similarity and Key BERT for keyword extraction to build detailed academic profiles and recommend relevant research articles or collaborators. The results demonstrated that traditional methods like TF-IDF rely mainly on word frequency failing to capture semantic meaning. The proposed combination of SBERT and KBERT performed the best in producing contextually accurate and semantically meaningful keywords [4].

The proposed system introduced Sentence-BERT (SBERT) which uses Siamese network structure to generate semantically meaningful sentence embeddings for similarity comparison. The study showed SBERT reduced the time required for similarity comparison from about 65 hours with standard BERT to roughly 5 seconds for larger datasets while maintaining high accuracy [5].

The researchers proposed system that uses a hierarchical cross-modal retrieval framework that learns from multi-level interactions between textual queries and video features to

improve text to video retrieval performance. The experiments on datasets of standard text-video retrieval showed higher retrieval effectiveness and better ranking metrics such as Recall@K, indicating more accuracy [6].

The proposed system introduces a fine-grained Cross-modal Alignment Network that aligns phrases in textual descriptions with detailed semantic units in videos to improve text-to-video retrieval accuracy. Experiments on datasets like YouCook2m and VATEX showed high retrieval effectiveness and improved ranking metrics proving fine-grained semantic alignments helps better contextual relations between video and text description [7].

The comprehensive survey of Neural Question Generation (NQG) methods that automatically generate questions from texts, knowledge bases using deep learning models. The survey analysed various architectures like RNN-based, Transformer-based, highlighting key evaluation metrics like BLEU, ROUGE and BERTcore, and identifies future research directions like multi-modal question generation and LLM-based methods [8].

The proposed system integrates Retrieval-Augmented Generation (RAG) with a digital interface to create an interactive AI tutor that provides personalised explanations for students in design education. Based on the experiments performed the students reported lower cognitive load during learning. The system demonstrated 96.2% factual accuracy in generated responses, showing the effectiveness of combining RAG model with digital human interaction for educational support [9].

The proposed system developed an educational chatbot using LLaMa-7B-Chat model combined with RAG to provide students with relevant resources for learning and accurate explanations. The system also used vector database using Pinecone API and embeddings with the all-MiniLM-L6-v2 model to access relevant educational materials before generating responses, resulting in accurate and context0aware answers compared to basic language models [10].

The comprehensive survey showed how LLMs are applied in document intelligence task such as information extraction and document-based question answering. It highlighted advancements for RAG, long-context processing and fine-tuning strategies for document centric understanding [11].

The researchers evaluated the ability of LLMs such as GPT-3.5 to generate summaries of medical evidence across multiple domains. The results showed the models could capture key information and generate summaries comparable in automatic metrics. However it was also found that automatic evaluation metrics did not strongly correlate with actual summary quality. This revealed the LLMs such as GPT models need for human assessment when using it in fields that are high-risk [12].

The proposed system developed an AI teaching assistant chatbot that compares traditional intent-based chatbots with an LLM based RAG model approach to provide better academic resource for students. The experimental results showed that RAG based LLM chatbot handled diverse student queries more efficiently than traditional systems, showing an overall average performance score of ≈ 4.76/5 [13].

III. Implementation

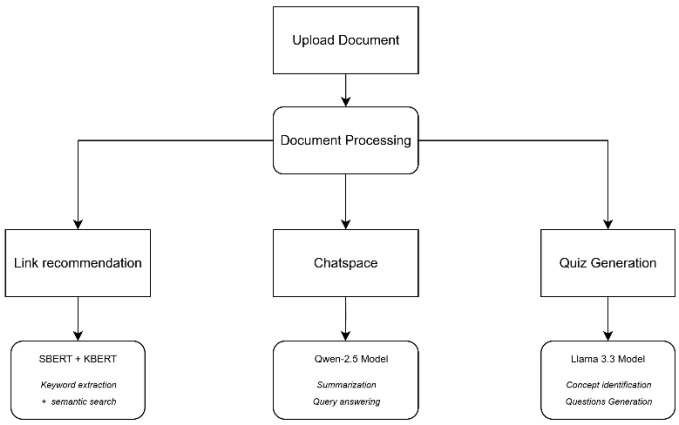

Fig. 1. Block diagram for proposed system

A. Upload Document

The user can upload the needed document after logging into the website. The proposed system allows user to upload in the format of PDFs, .docx, .txt files.

B. Document Processing

The Document Processing module converts the uploaded documents to machine readable text using OCR techniques (for images) and parses directly for PDF files. Processing of the document helps remove noise and normalises the content which significantly improves the performance [12] of NLP tasks like summarization and query answering tasks further in the system.

C. Link recommendation

The recommendation system uses a pre-trained SBERT model, all-MiniLM-L6-v2, from SentenceTransformers framework, used to generate vector embeddings of the documents for semantic similarity [5]. The KBERT extracts the top keywords from the normalised text. These resources are then ranked again using cosine similarity between the embeddings.

$$Similarity = \cos(\theta) = \frac{(A \cdot B)}{(||A||||B||)}$$

this ensures contextually relevant recommendations. External APIs like YouTube Data API for YouTube videos link recommendations as well as custom google search API for scholarly search results of articles/ websites are used.

D. Chatspace

The chat space section enables the user to ask query about the document content and can also provide with the summarization of the content. The chat space module uses the Qwen-2.5 LLM for summarization and context-based query answering. Qwen models demonstrate effectiveness in generating content and context aware responses while LLM based conversational systems show enhanced engagement

with the users which can lead to personalised learning experiences [13].

E. *Quiz Generation*

The quiz generation module helps promote active learning and provides strictly content-based questions to test the knowledge of the user regarding the said topic. This module utilizes the Llama 3.3 model, a deep learning model that is effective for automatically generating meaningful questions from textual input [8], to identify the key concepts based on the content extracted from the document. The model processes the text into structured MCQ style questions. It also scores the user based on the correctness of the answers chosen.

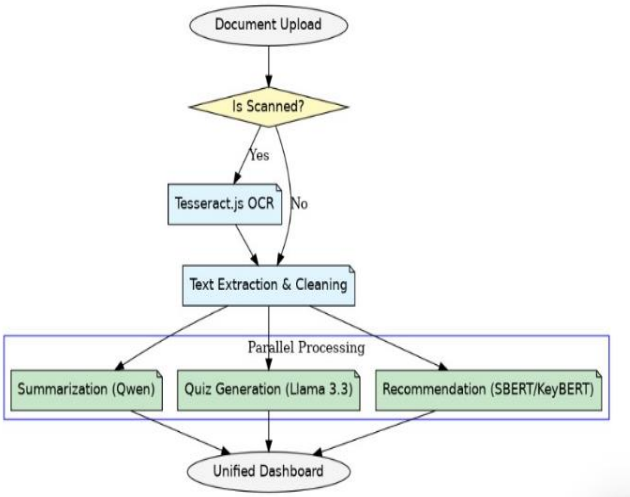

Fig. 2. Flowchart of the proposed system

IV. Results and Evaluation

The system was evaluated using multiple academic documents to check the system's accuracy, running time and usability. Total 10 documents were uploaded for evaluation. Out of which 7 documents were of long formed academic content and the rest were short formed extra-curricular content. The documents were used to evaluate the summarization module, articles and YouTube links extraction module and quiz module respectively.

The summarization module gave strong and concise summary based on the whole documents. And the module also provided correct answer to the prompts based on the document. The reference module provided relevant and recent articles and YouTube links based on the entire document. This results in better user efficiency as the module helps the user to explore the topics beyond the uploaded document. As for the quiz module, the generated question where strongly based on the entire document also including important topics. The module can generate different set of questions for the same document providing better efficiency. However, the system supports only DOCX documents for quiz generation, which represents an opportunity for future improvement.

We compared our system with ChatGPT to measure the system's accuracy and efficiency.

Table 1: Comparative workflow analysis between Standalone AI tool (ChatGPT) and the Proposed system

Metric	ChatGPT	Proposed system
Time to result	Approximately 5 minutes	Less than 1 minute
Multimedia support	Text only	Text, Articles and YouTube links
Quiz generation	None	Automated quiz based on the document
Workflow integration	Separate tasks	All-in-one system
Context-switching required	Medium (2-3 tools)	None (single platform)

This comparison highlights the proposed system's dominance over basic AI tools such as ChatGPT. While ChatGPT provides powerful text-based results, the entire system is dependent on the prompts from the user. This results in increase in running time as the context-switching time is more. On the other hand, the proposed system integrated all the modules into a single interface resulting in less task completion time. The results demonstrate that the proposed system offers a more user-friendly and efficient workflow for document-based learning.

V. Conclusion and Future Scope

The future scope for this system includes focusing on building a vector database-based RAG system that will help handle large scale documents and diversify the kind of material resources for learning. Implementation of multilingual options to help expand and make the system accessible for a wider range of students, personalized user analytics to help students keep track of completed topics and to hold record of their scores leading to recommendation of topics to learn for the student focusing on their weaker topics. The proposed system demonstrates the successful development of a unified AI study assistant that effectively bridges the gap between document ingestion and active knowledge synthesis by integrating multiple specialized AI components including the Qwen model for summarization, Llama 3.3 for pedagogical assessment, and Sentence Transformers library for semantic recommendation. The platform allows users to maintain a state of flow and focus their mental energy on comprehension rather than tool orchestration.

References

[1] M. Odabaşı, M. Yaşar, and B. Öztürk, "Benchmarking large language models for text summarization: A comparative study of GPT, Qwen, and Llama models," *arXiv preprint*, arXiv:2501.xxxxx, 2025.

[2] Y. Chen, L. Wang, and H. Zhang, "Prompt learning with Qwen for domain-specific document summarization," in *Proc. Int. Conf. Natural Language Processing (ICNLP)*, 2025, pp. 234–241.

[3] J. Liu, X. Wang, and S. Chen, "Phrase-level representations for content-based article recommendation," in *Proc. ACM Conf. Recommender Systems (RecSys)*, 2024, pp. 156–164.

[4] C. Ayala-Tipán, L. Recalde, and G. Suntaxi, “Enhancing academic profiling with advanced NLP techniques: SBERT + KeyBERT,” in Communications in Computer and Information Science, vol. 2496, pp. 195–211, 2025.

[5] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, 2019, pp. 3982–3992.

[6] J. Jiang, S. Min, W. Kong, D. Gong, H. Wang, Z. Li, and W. Liu, “Tencent text-video retrieval: Hierarchical cross-modal interactions with multi-level representations,” IEEE Access, vol. 10, pp. 12315–12327, 2022.

[7] N. Han, J. Chen, G. Xiao, H. Zhang, Y. Zeng, and H. Chen, “Fine-grained cross-modal alignment network for text-video retrieval,” in Proc. ACM Int. Conf. Multimedia Retrieval (ICMR), 2021, pp. 492–500

[8] S. Guo, L. Liao, C. Li, and T.-S. Chua, “A survey on neural question generation: Methods, applications, and prospects,” in Proc. 33rd Int. Joint Conf. Artificial Intelligence (IJCAI), 2024, pp. 8038–8047.

[9] X. Zhou, S. Zhao, P. Wu, and Y. Chen, “System design and evaluation of RAG-enhanced digital humans in design education: Analyzing cognitive load and instructional efficiency,” Applied Sciences, vol. 16, no. 2, p. 1068, 2026.

[10] D. S. Cabezas, R. Fonseca-Delgado, I. Reyes-Chacón, P. Vizcaino-Imacaña, and M. E. Morocho-Cayamcela, “Integrating a LLaMa-based chatbot with augmented retrieval generation as a complementary educational tool for high school and college students,” in Proc. 19th Int. Conf. Software Technologies (ICSOFT), 2024, pp. 395–402.

[11] W. Ke et al., "Large Language Models in Document Intelligence: A Comprehensive Survey, Recent Advances, Challenges and Future Trends," ACM Trans. Inf. Syst., vol. 43, no. 4, 2025.

[12] L. Tang et al., "Evaluating large language models on medical evidence summarization," npj Digit. Med., vol. 6, no. 1, p. 158, 2023. DOI: 10.1038/s41746-023-00896-7.

[13] B. Alsafari, E. Atwell, A. Walker, and M. Callaghan, “Towards effective teaching assistants: From intent-based chatbots to LLM-powered teaching assistants,” *Natural Language Processing Journal*, vol. 8, p. 100101, 2024..